

Final Analyzed Data: realwaste-main

Unit of Observation: Each observation represents a single item of waste collected at the point of reception. Each observation is associated with a material type, such as plastic, or vegetation.

Total images: 4752

Variables

Image: This variable consists of the visual pixel data representing a piece of waste photographed on a concrete floor in a landfill environment

- **Source & Processing:** The original images were provided at a resolution of 524x524 pixels. All inputs were resized to a consistent dimension and normalized by rescaling pixel values to a standard range to ensure model stability. To simulate real-world conditions, training data underwent augmentation including random horizontal/vertical flips, rotations, and color jitter.
- **Observations:** 4752(0)
- **Data Type:** Image

Material Type: This categorical variable identifies the material type of the waste item into one of nine distinct disposal categories: Cardboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textile Trash, or Vegetation.

- **Source & Processing:** The dataset was downloaded in a pre-sorted format where images were organized into folders. The folder names were extracted and used as the label for each observation. A weighted cross-entropy loss was later applied during modeling to address class imbalances between these categories
- **Observations:** 4752(0)
- **Frequency table:**

Category	Number of images
Plastic	921
Metal	790
Paper	500
Miscellaneous Trash	495
Cardboard	461
Vegetation	436

Glass	420
Food Organics	411
Textile Trash	318

- **Distribution:** The distribution of categories is imbalanced, with Plastic being the most frequent material type and Textile Trash being the least represented